

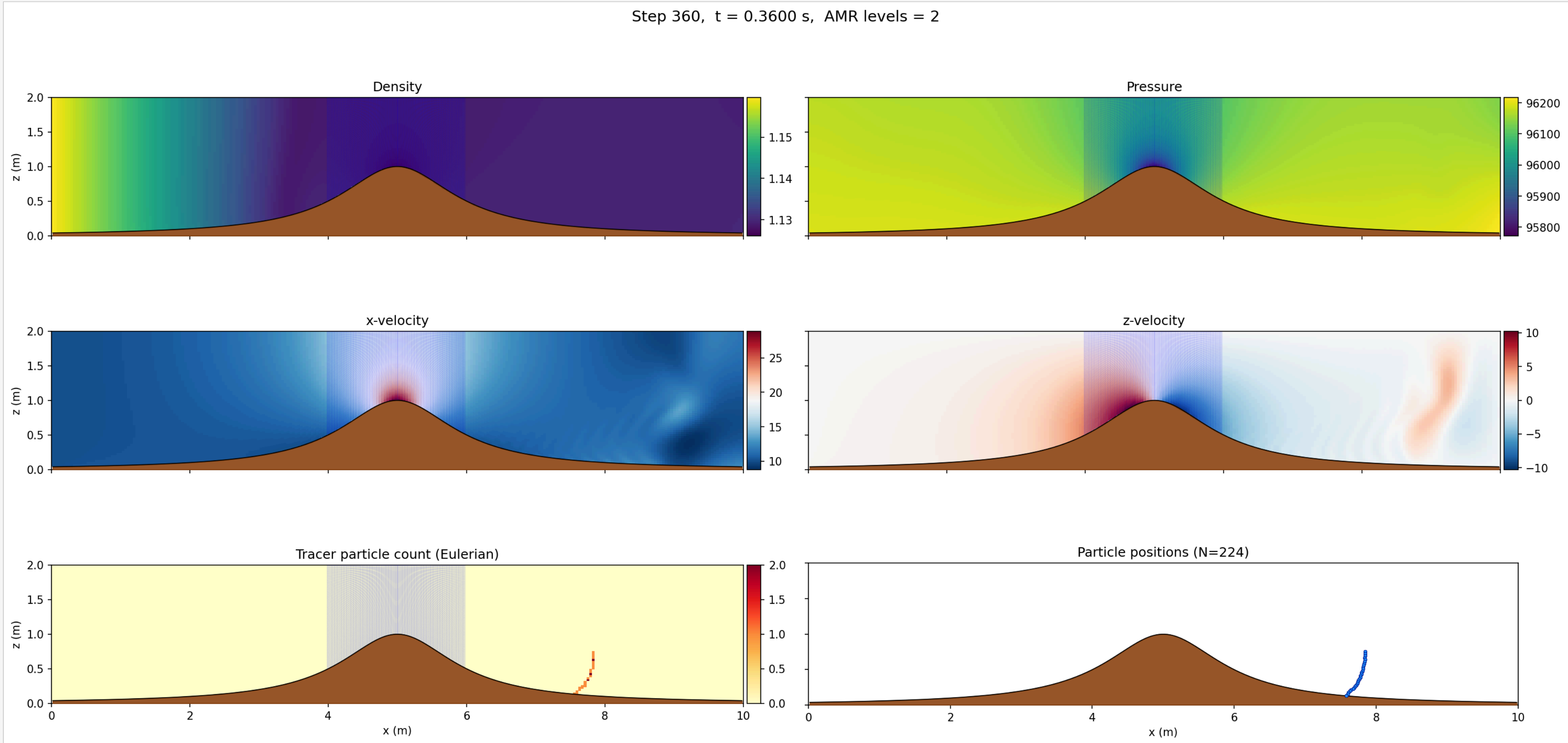
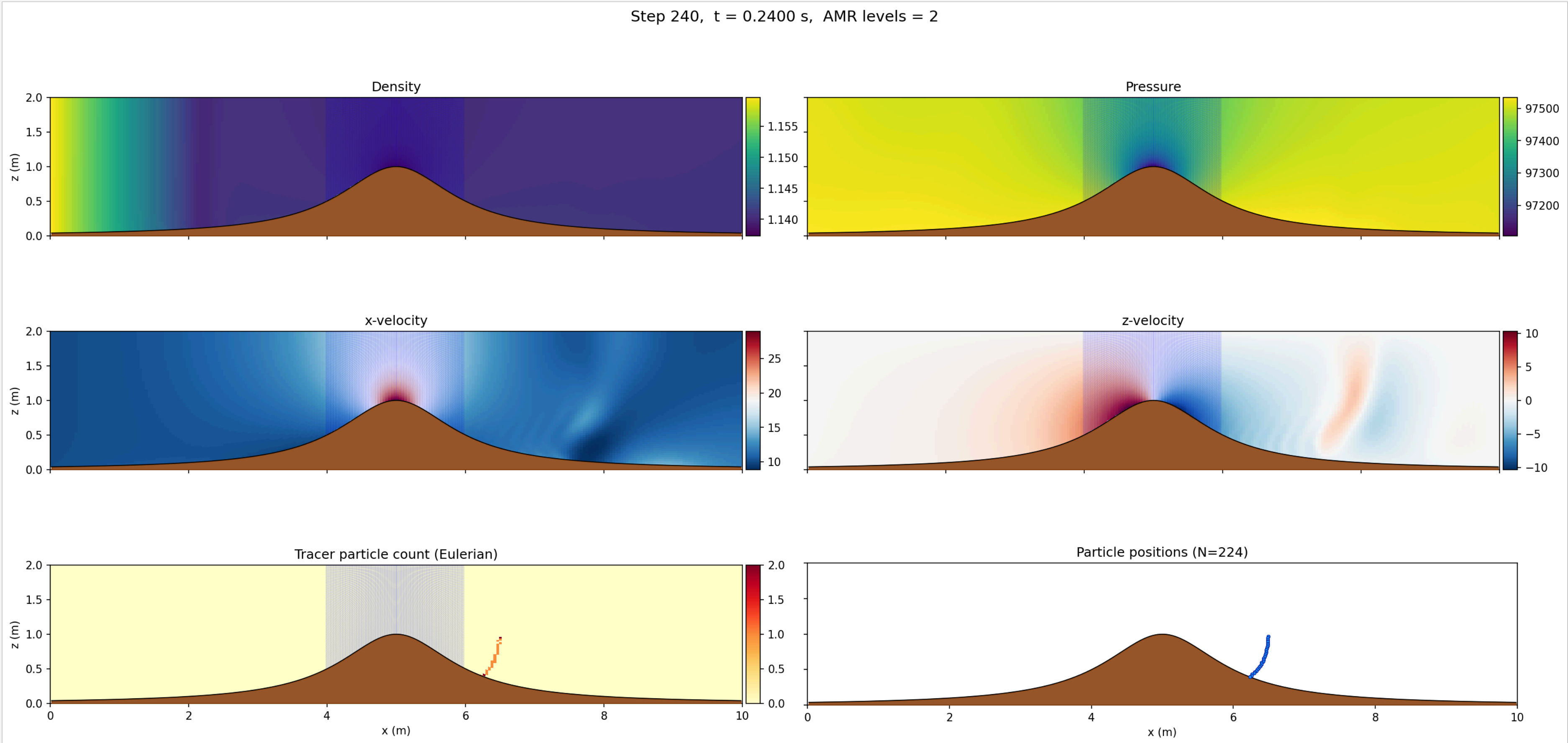
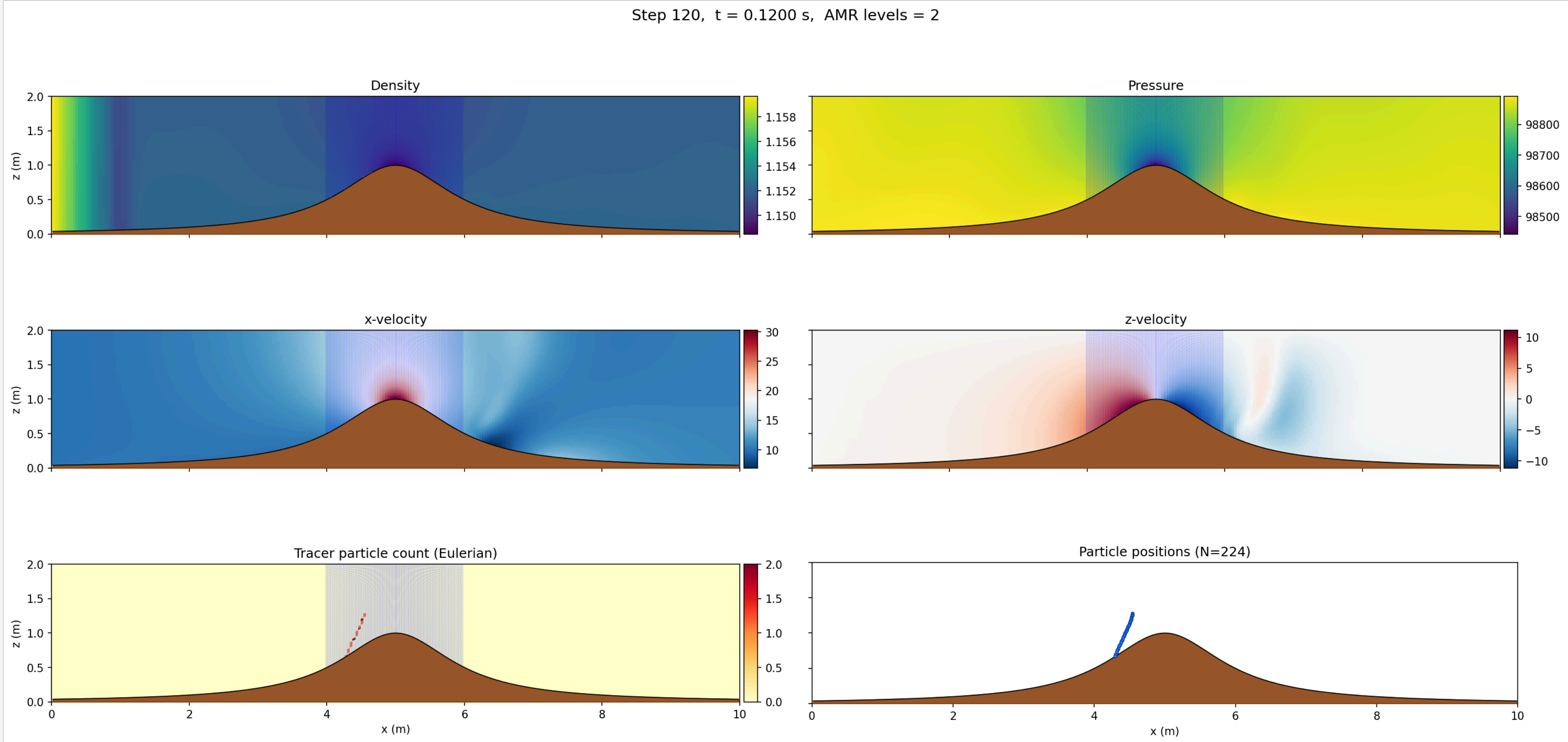
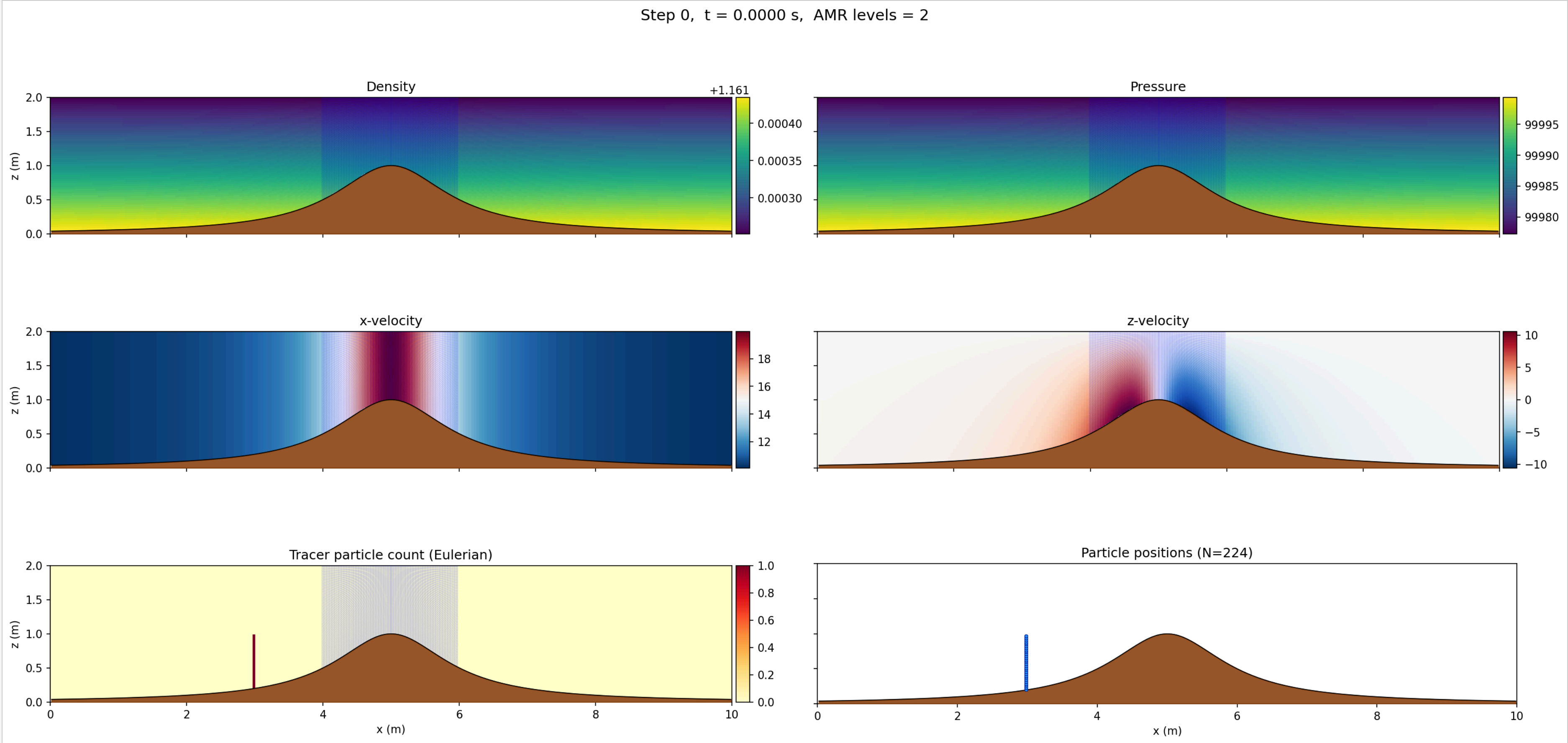
One-Way Coupling: Failed Cases (2026-04-08)

Branch: dg/particles_w_AMR — 6 of 7 terrain AMR cases failed (all flat-terrain cases pass)
All failures are due to incorrect coarse-fine interpolation in terrain-following coordinates.

1. Terrain (WoA), AMR1, box_fullz

Input: `inputs_over_hill_AMR1_box_fullz`
Negative density at coarse step 373 on L1. Static box covers full z-extent with 1 refined level (ratio 2,1,2). Unlike two-way coupling where this case passed (averaging-down partially stabilized the solution), one-way coupling lacks that feedback and the instability grows until density goes negative at cell (304, 0, 0) with $\rho = -0.346$.

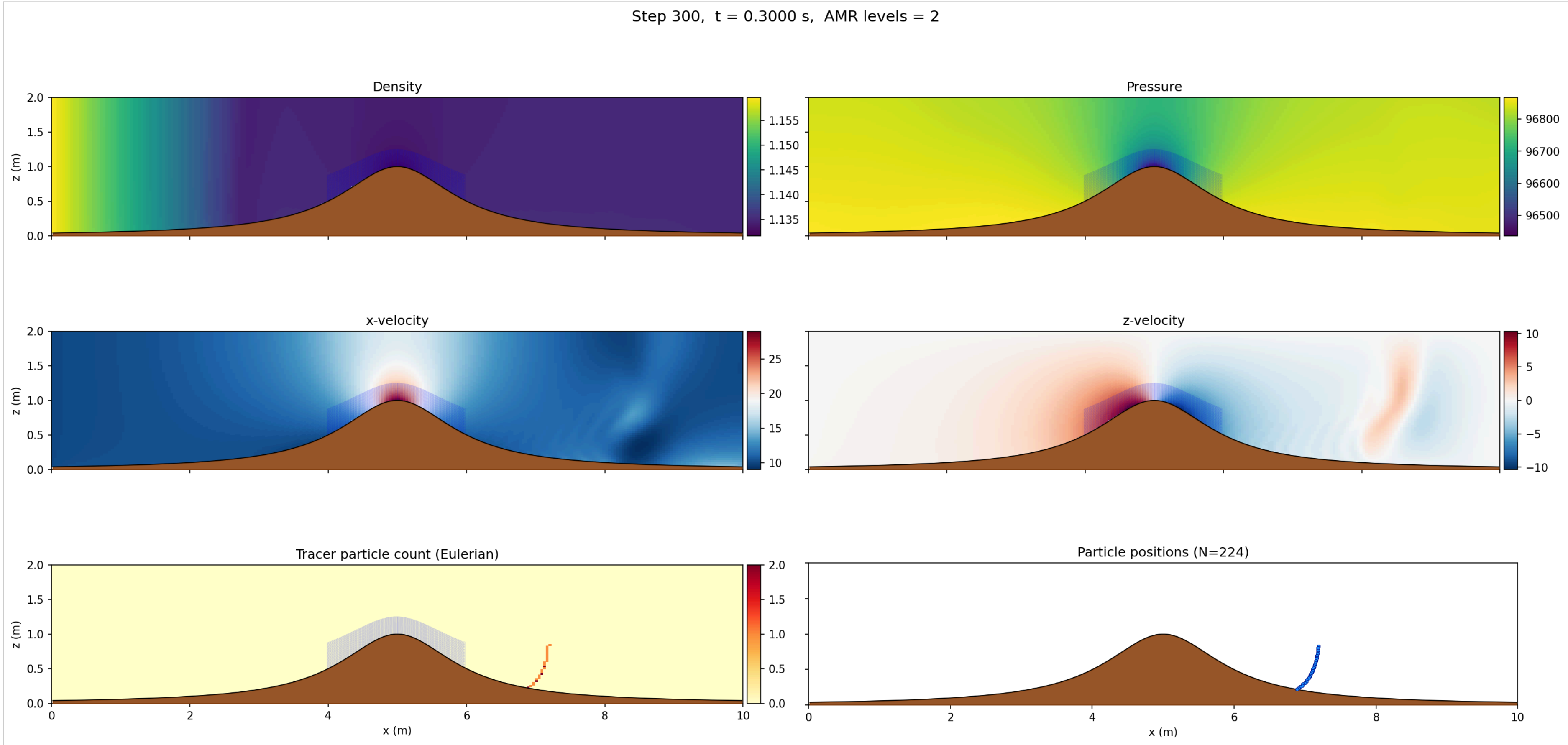
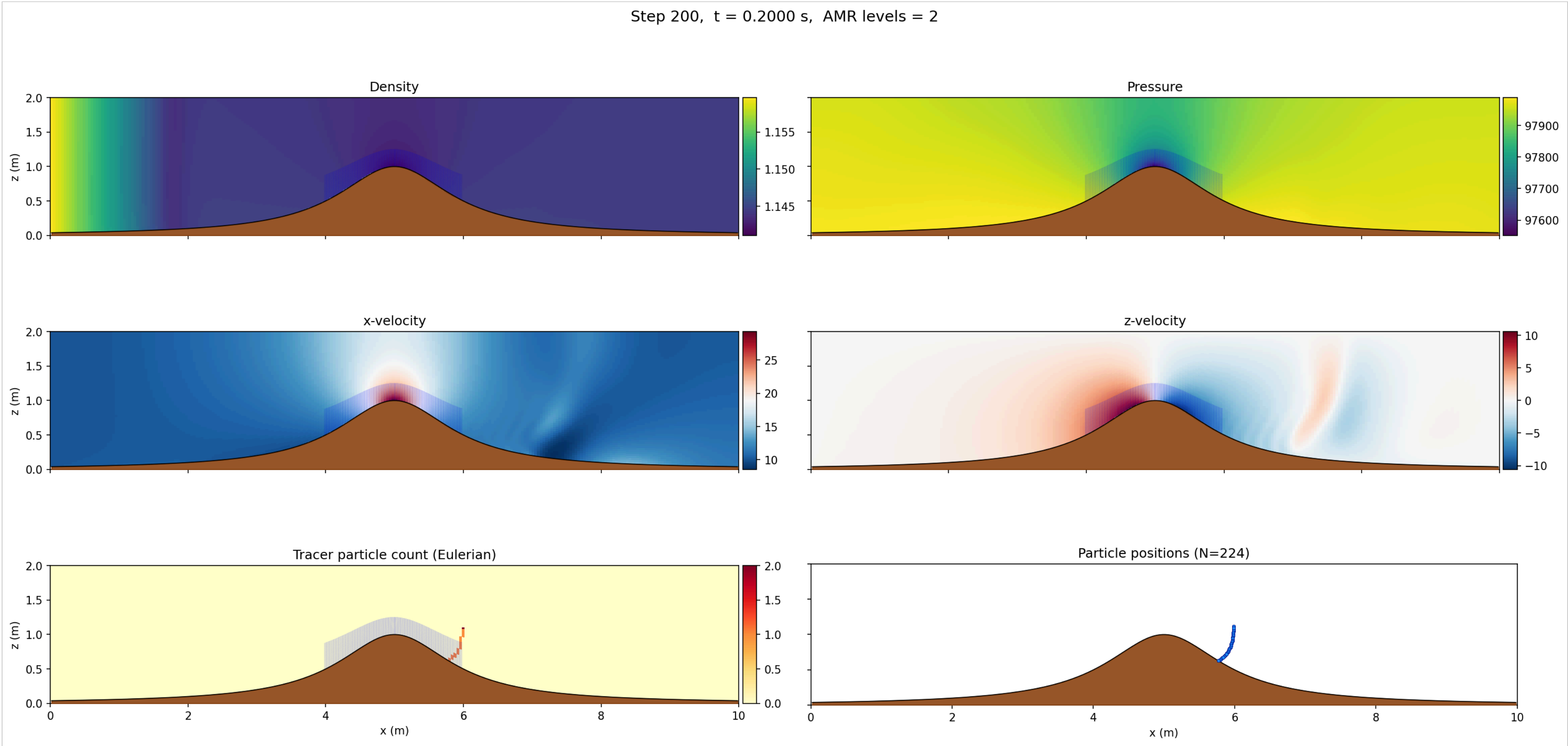
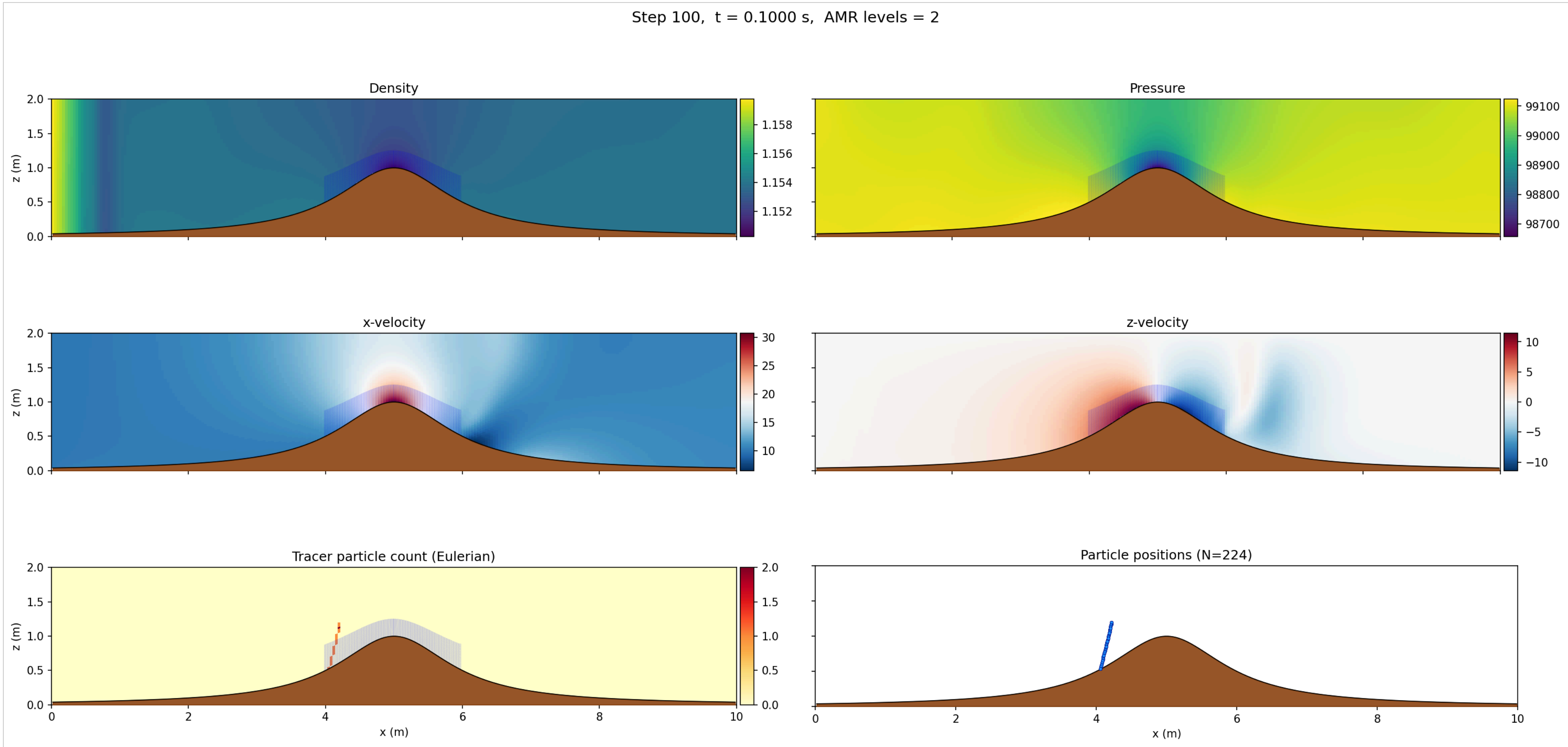
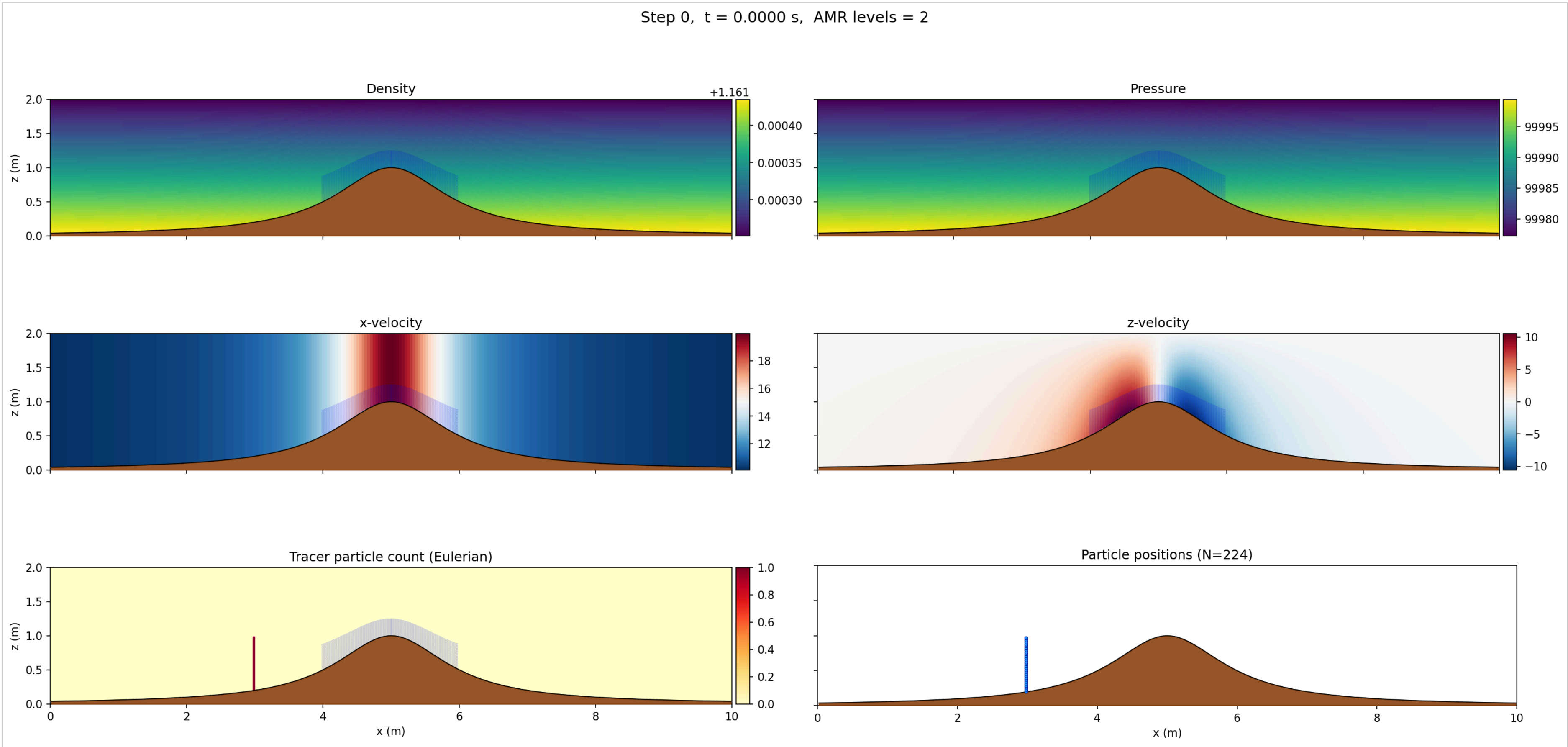
Backtrace: `amrex::Abort` called from `ERF::check_for_negative_theta` (`ERF.cpp:3237`) after `ERF::Advance` (`ERF_Advance.cpp:269`). Message: "Bad rho in check_for_negative_theta".



2. Terrain (WoA), AMR1, box_partialz

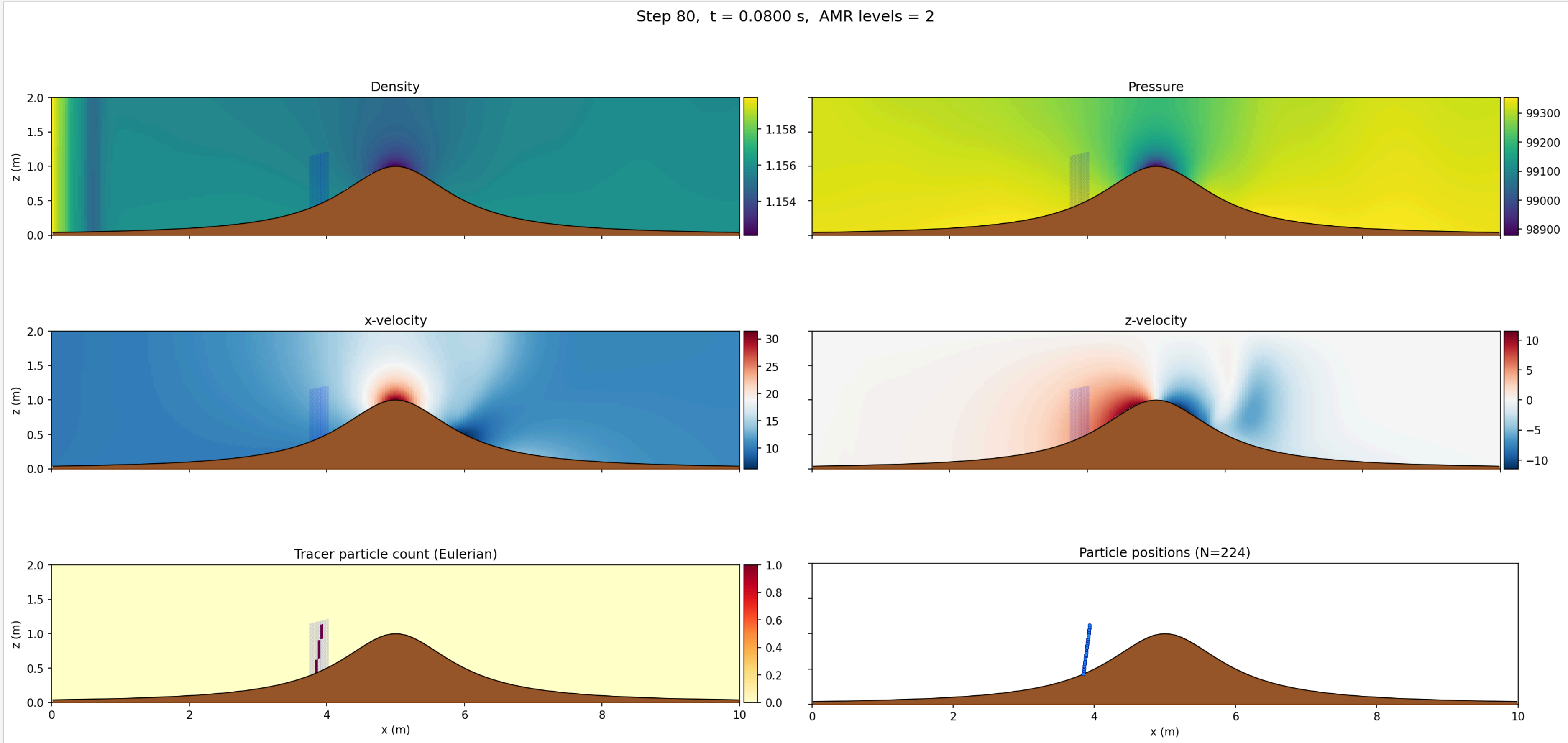
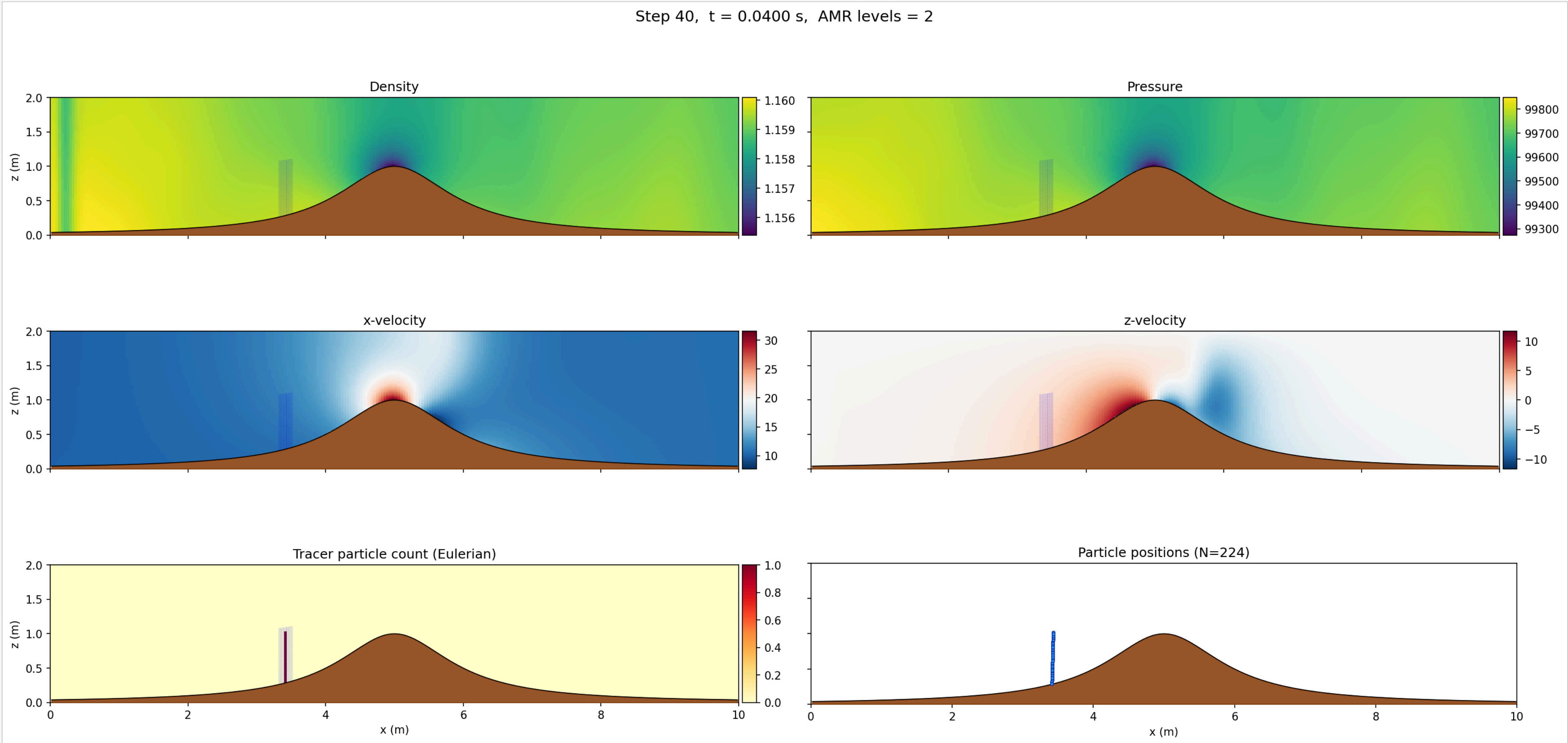
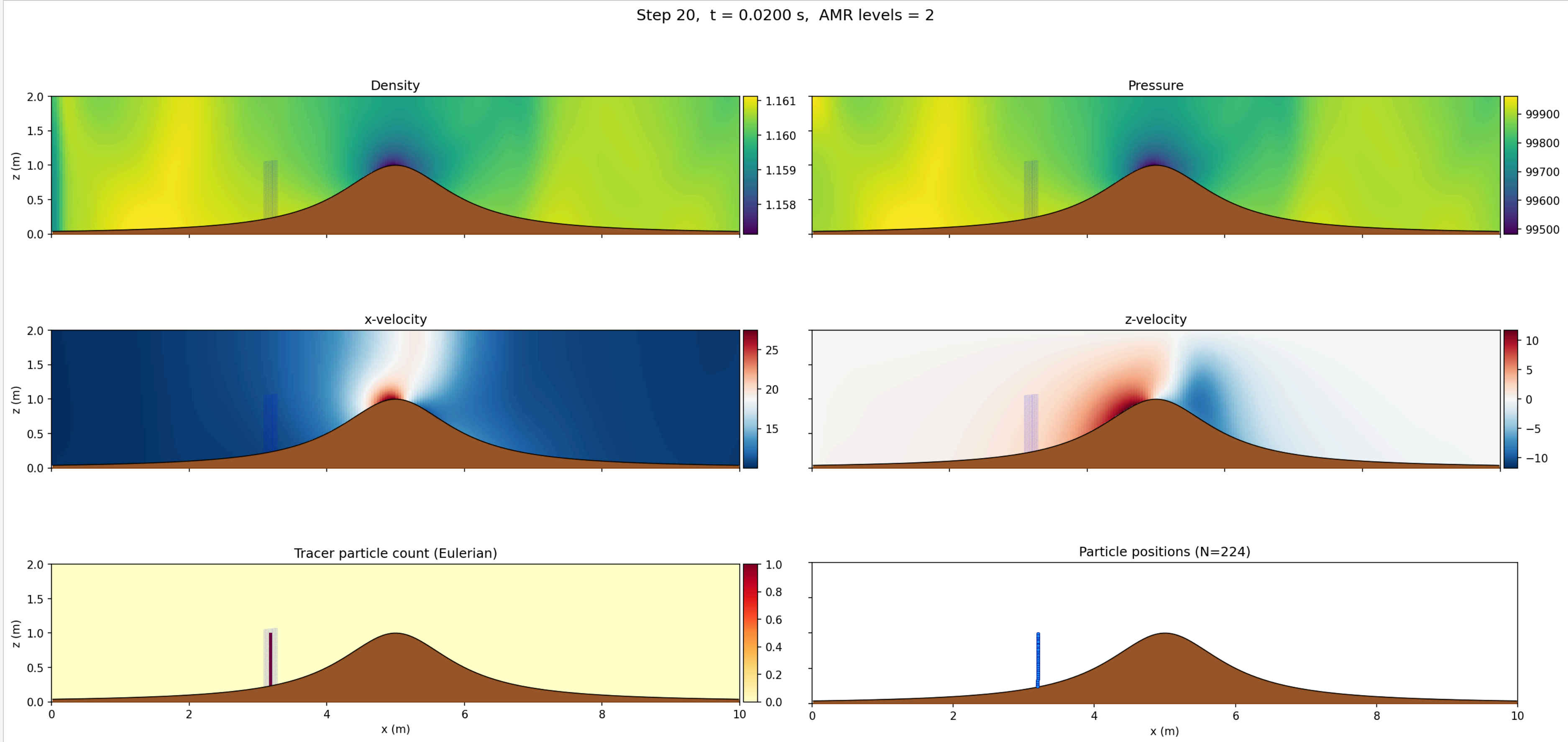
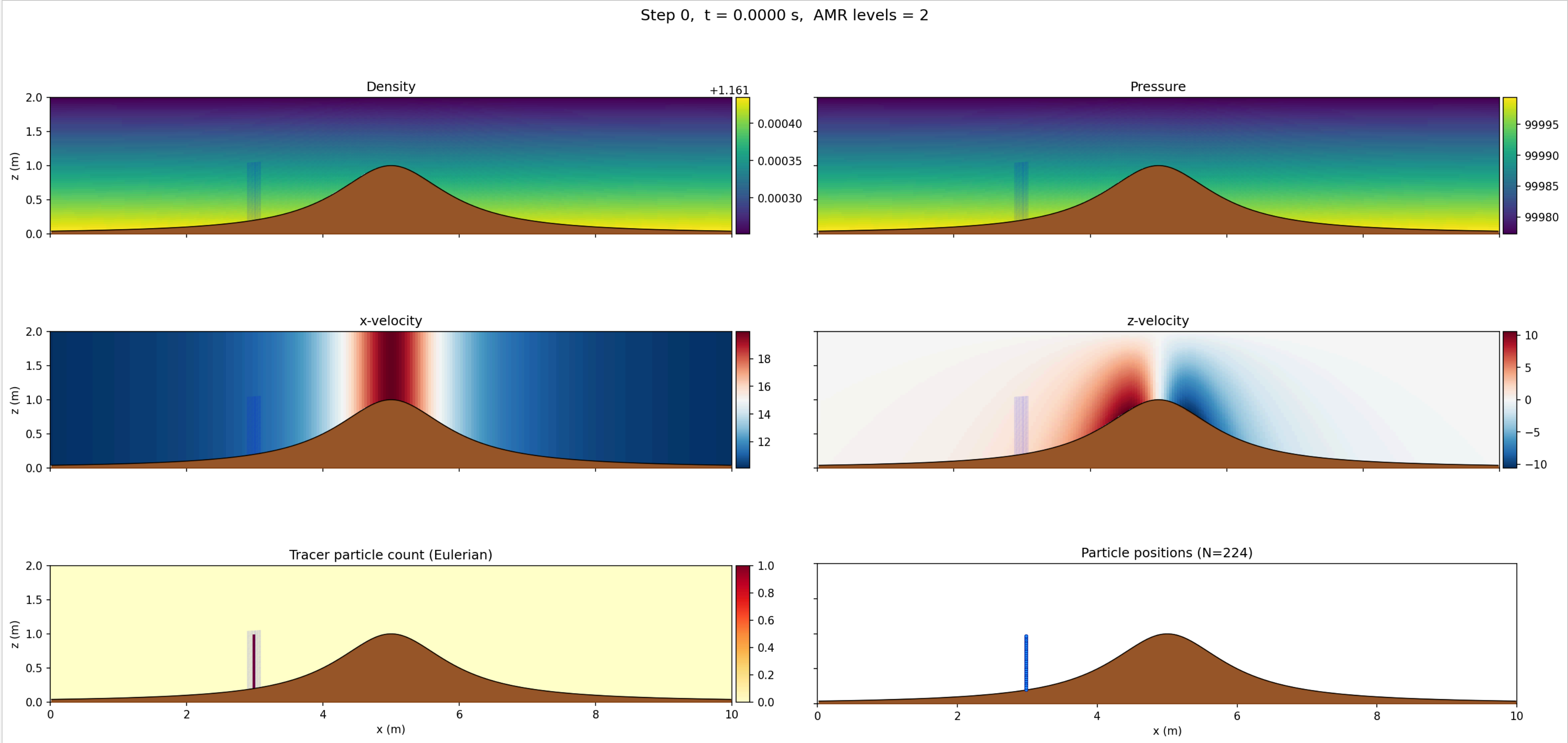
Input: [inputs_over_hill_AMR1_box_partialz](#)
FPE (floating-point exception) at coarse step 305 during L1 time integration. Static box covers partial z-extent with terrain-following coordinates. The index-space interpolation at the top of the refined region assumes uniform z-spacing, producing incorrect values that grow into the flow field.

Backtrace: signal caught in [powf64](#) (libm) called from [ERF_TI_utils.H:41](#) inside the slow RHS lambda of [advance_dycore](#) → [ERF_TI_slow_rhs_pre.H:26](#) . The FPE occurs when NaN/Inf values from the C-F boundary reach the exponentiation in the RHS computation.



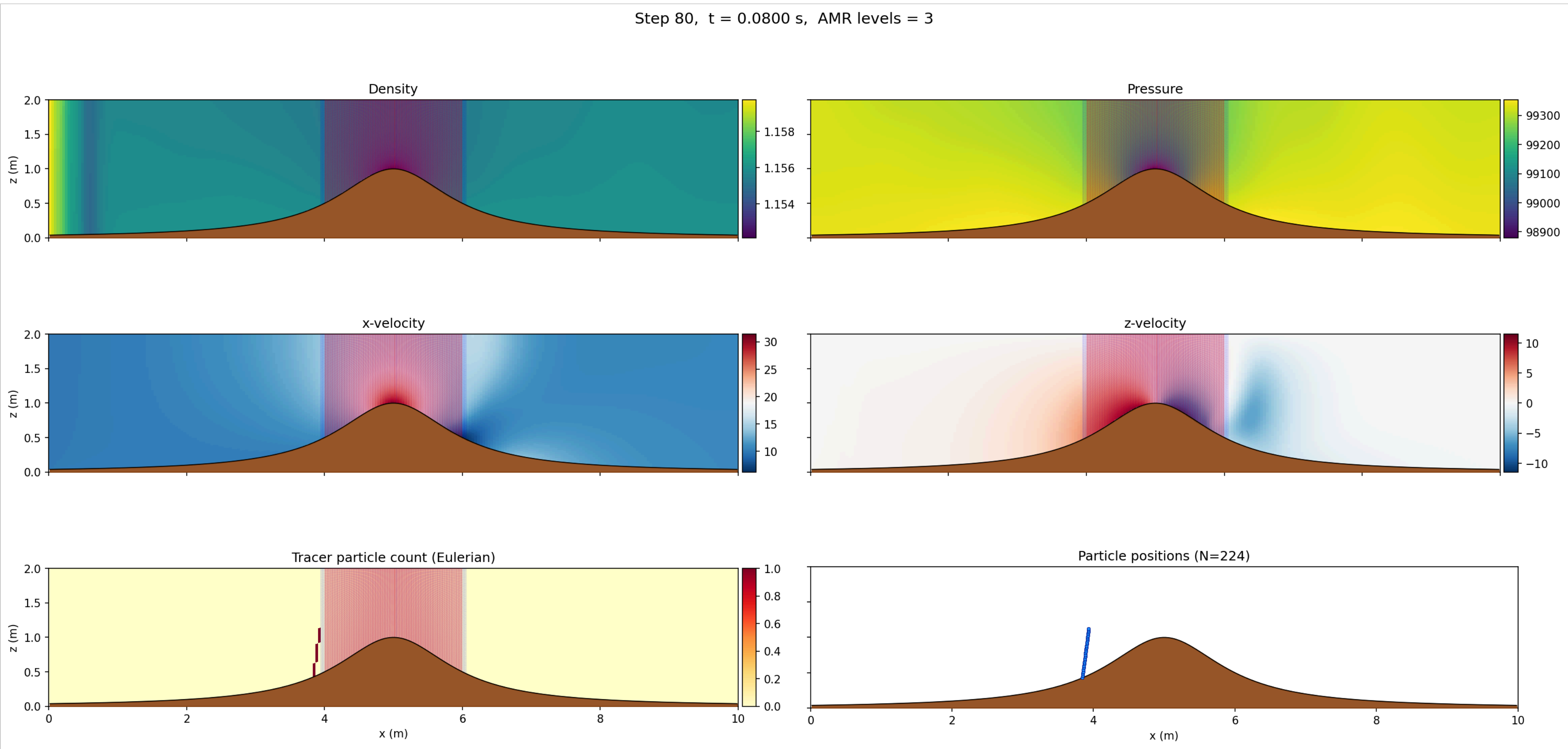
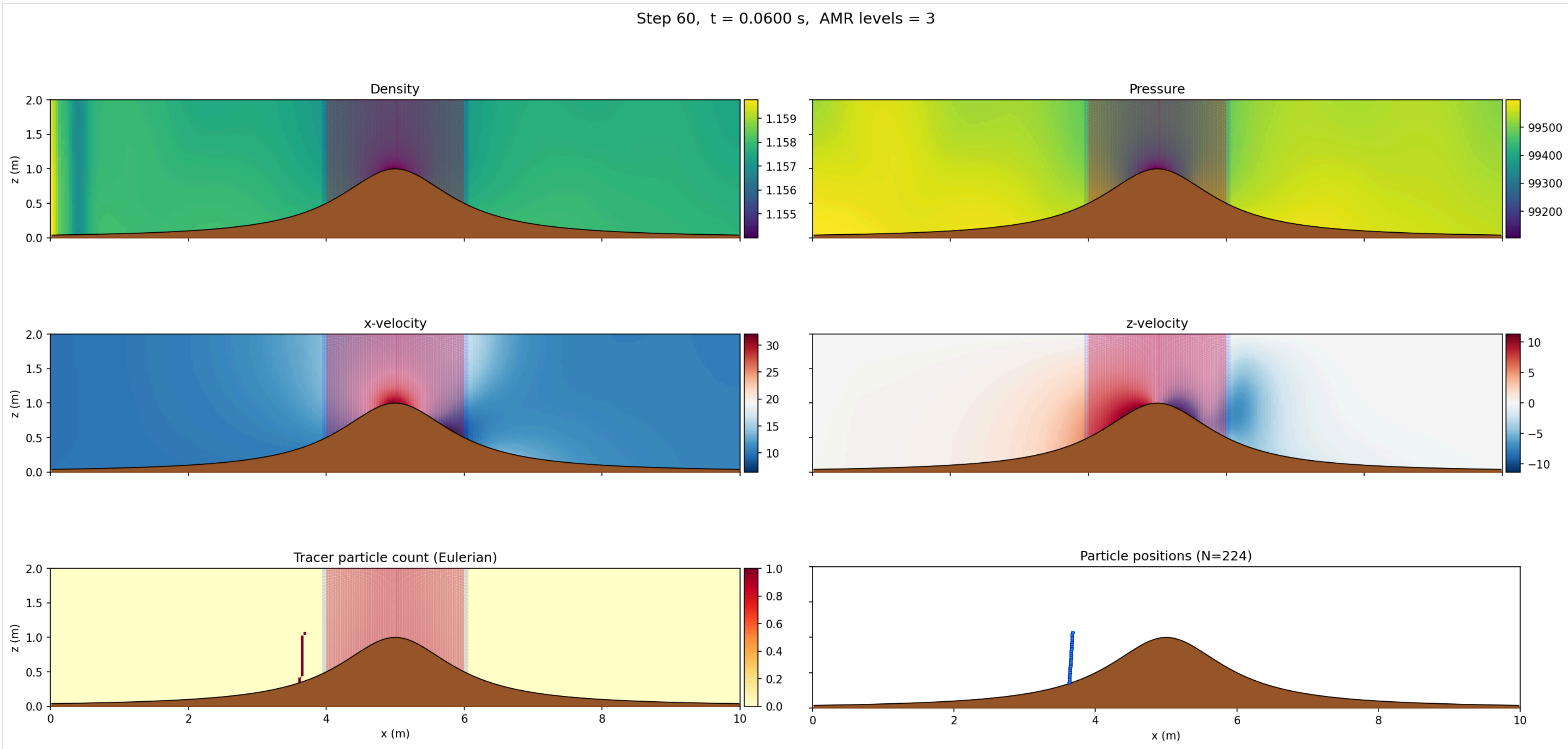
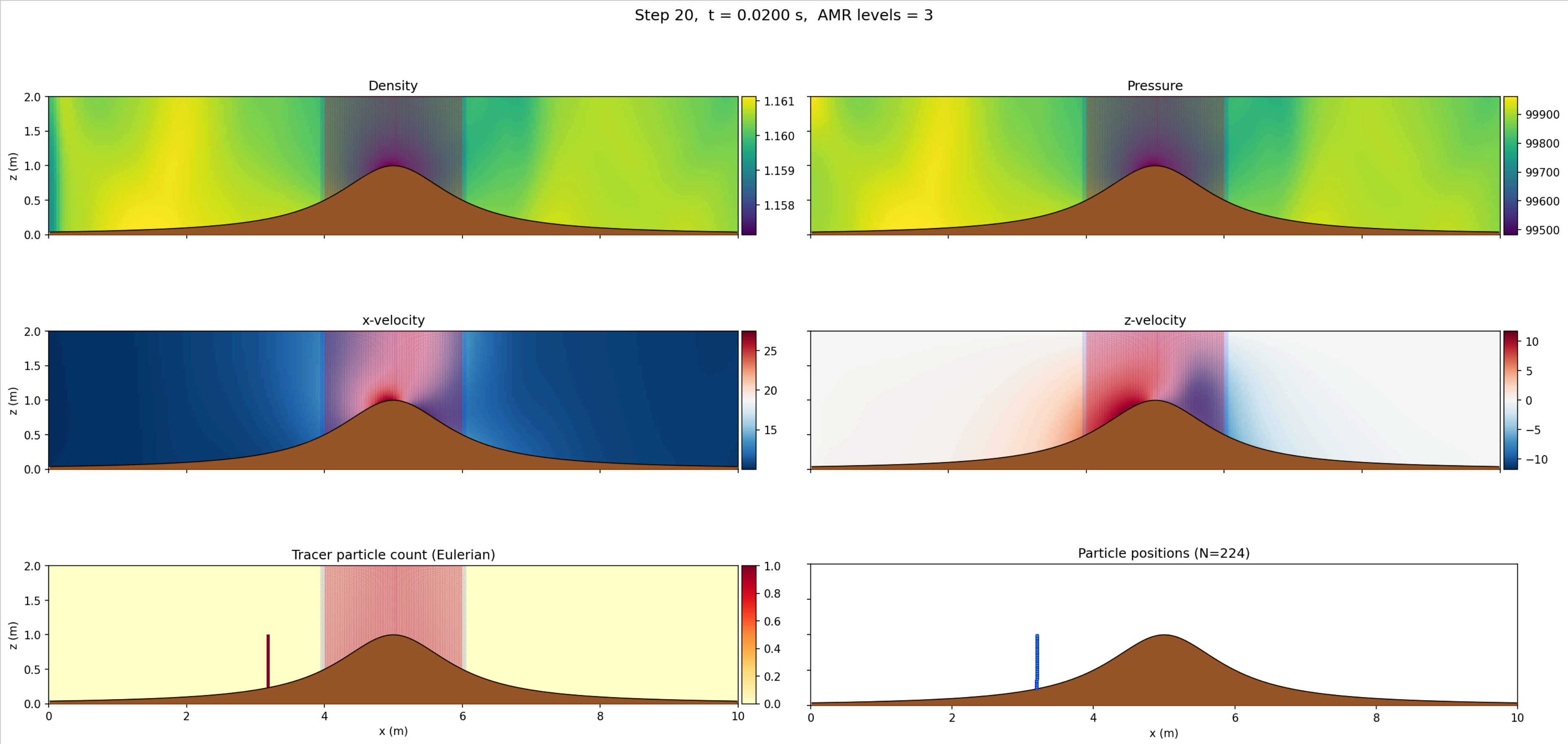
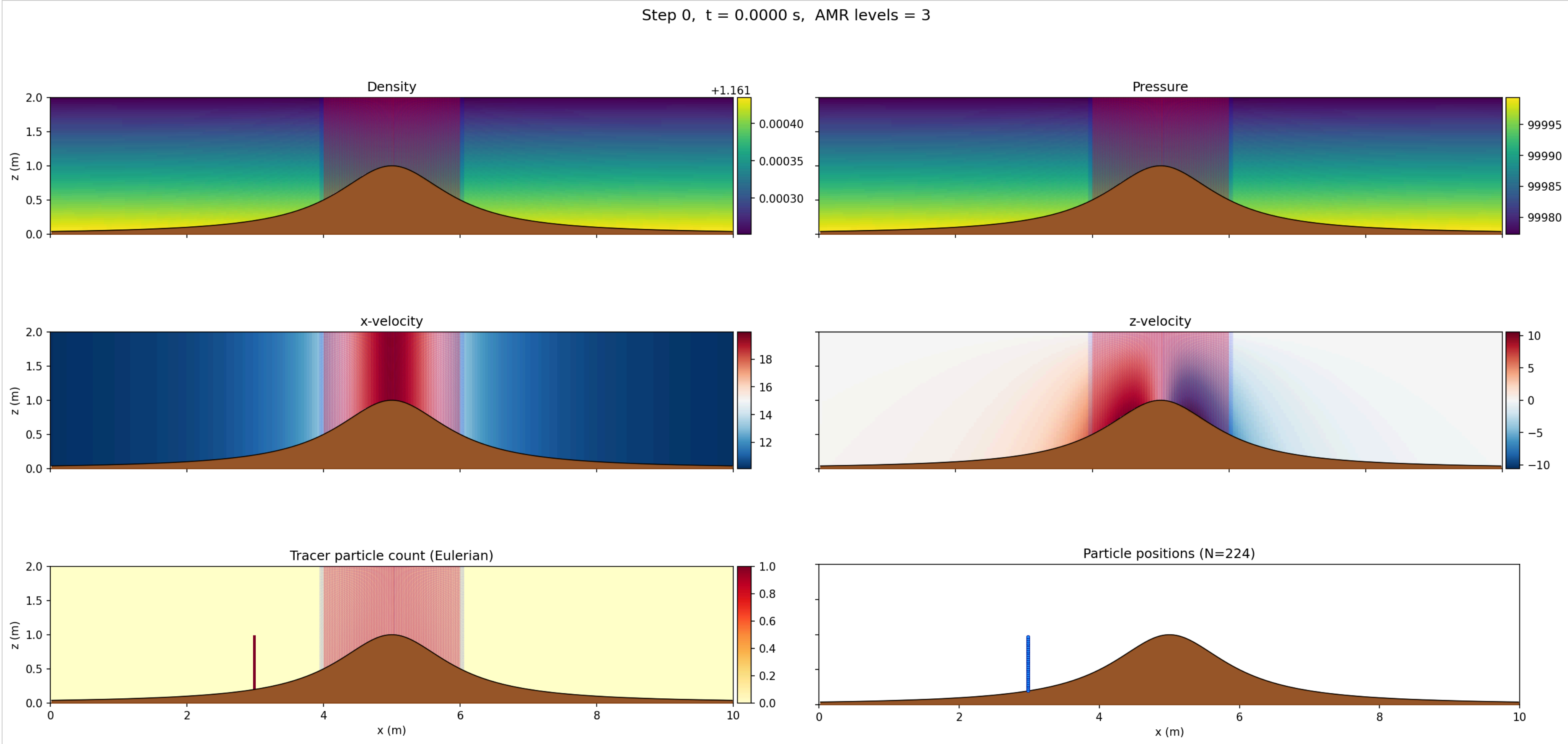
3. Terrain (WoA), AMR1, particlecount

Input: [inputs_over_hill_AMR1_particlecount](#)
Memory corruption ("corrupted size vs. prev_size") at coarse step 100. Dynamic particle-count tagging with terrain-following coordinates. The refined region moves with the particles, and the coarse-fine interpolation for newly created fine cells uses incorrect z-offsets.
Backtrace: SIGABRT from heap corruption. The heap was corrupted by prior bad data from terrain C-F interpolation; the crash manifests during memory operations after the flow state has already been corrupted by incorrect coarse-fine interpolation in terrain-following coordinates.



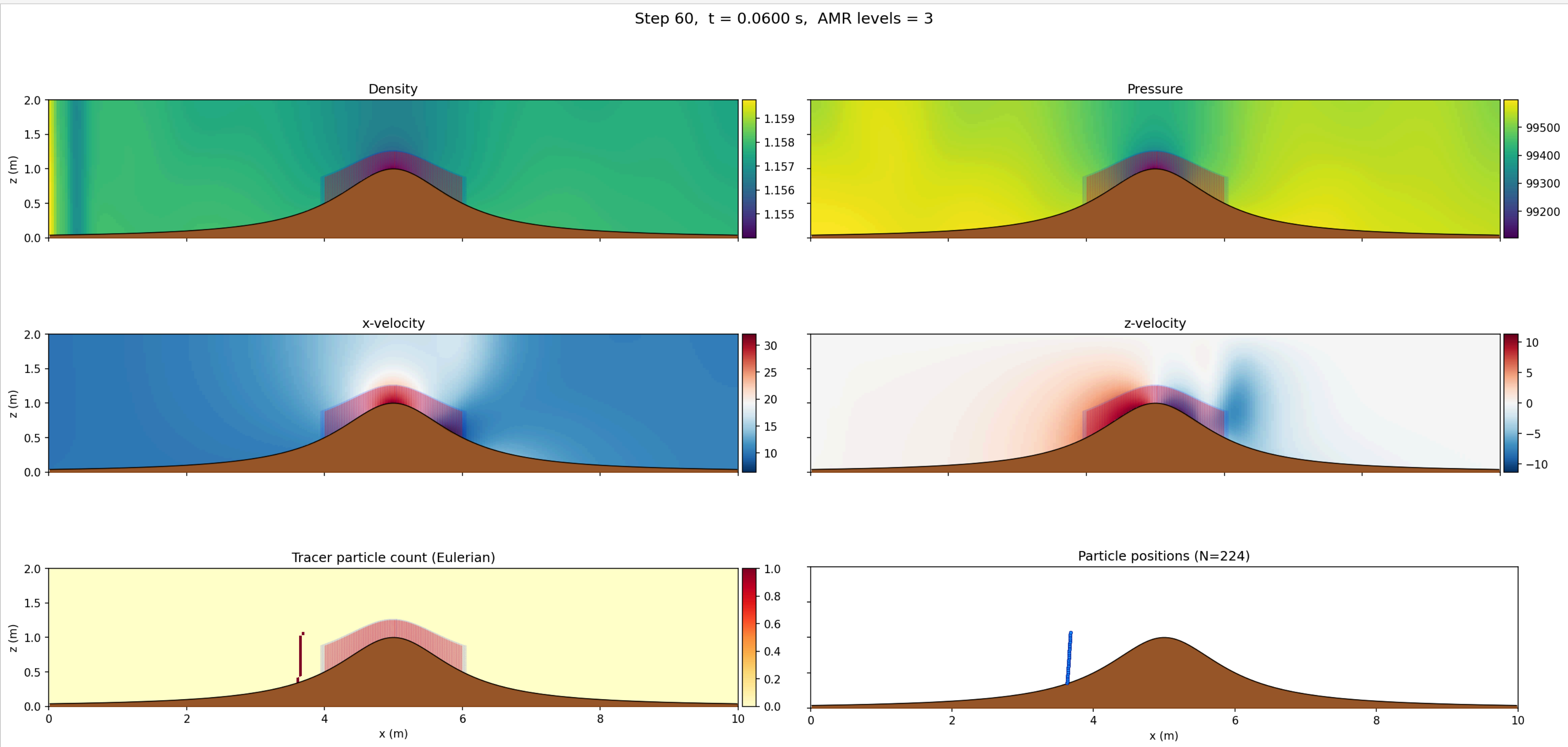
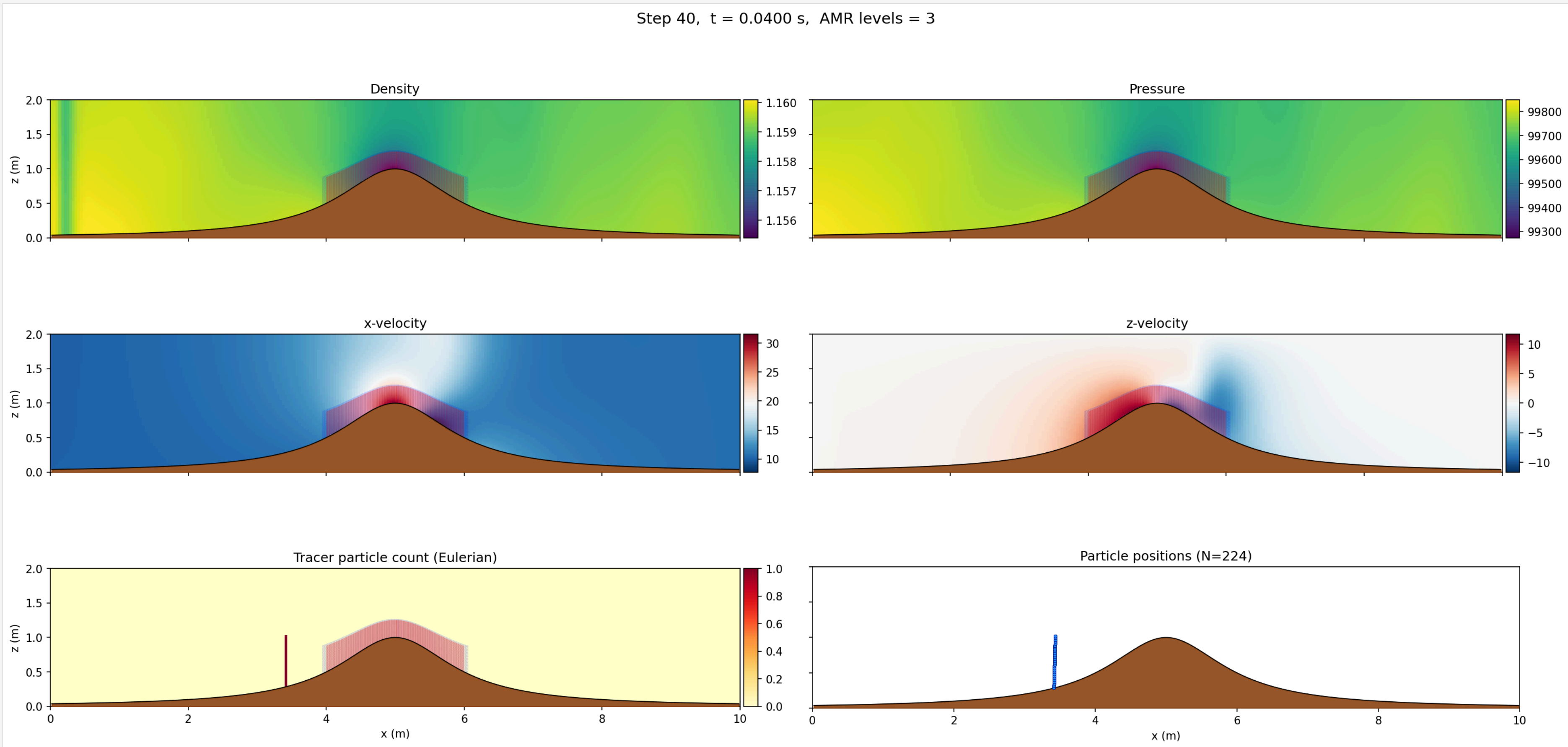
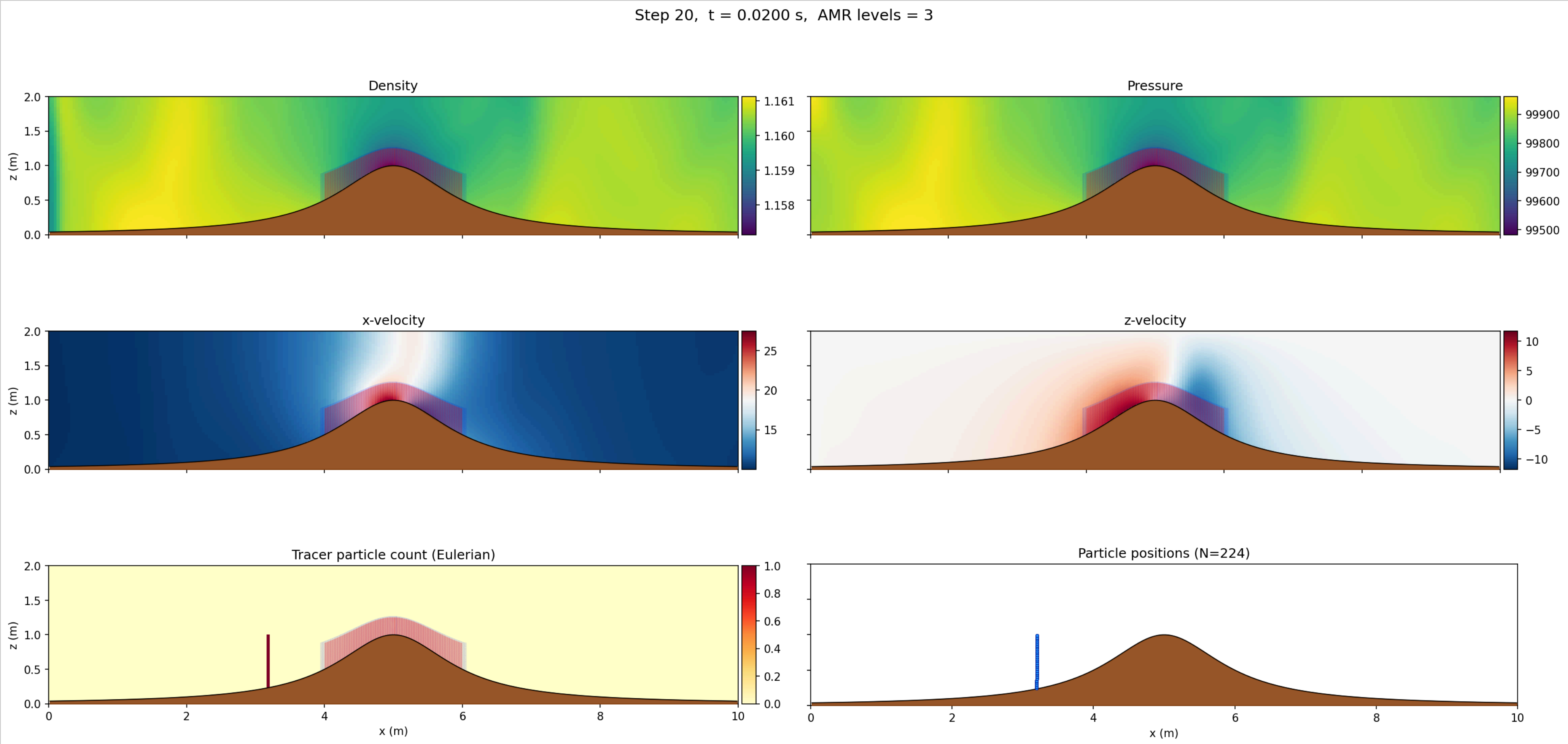
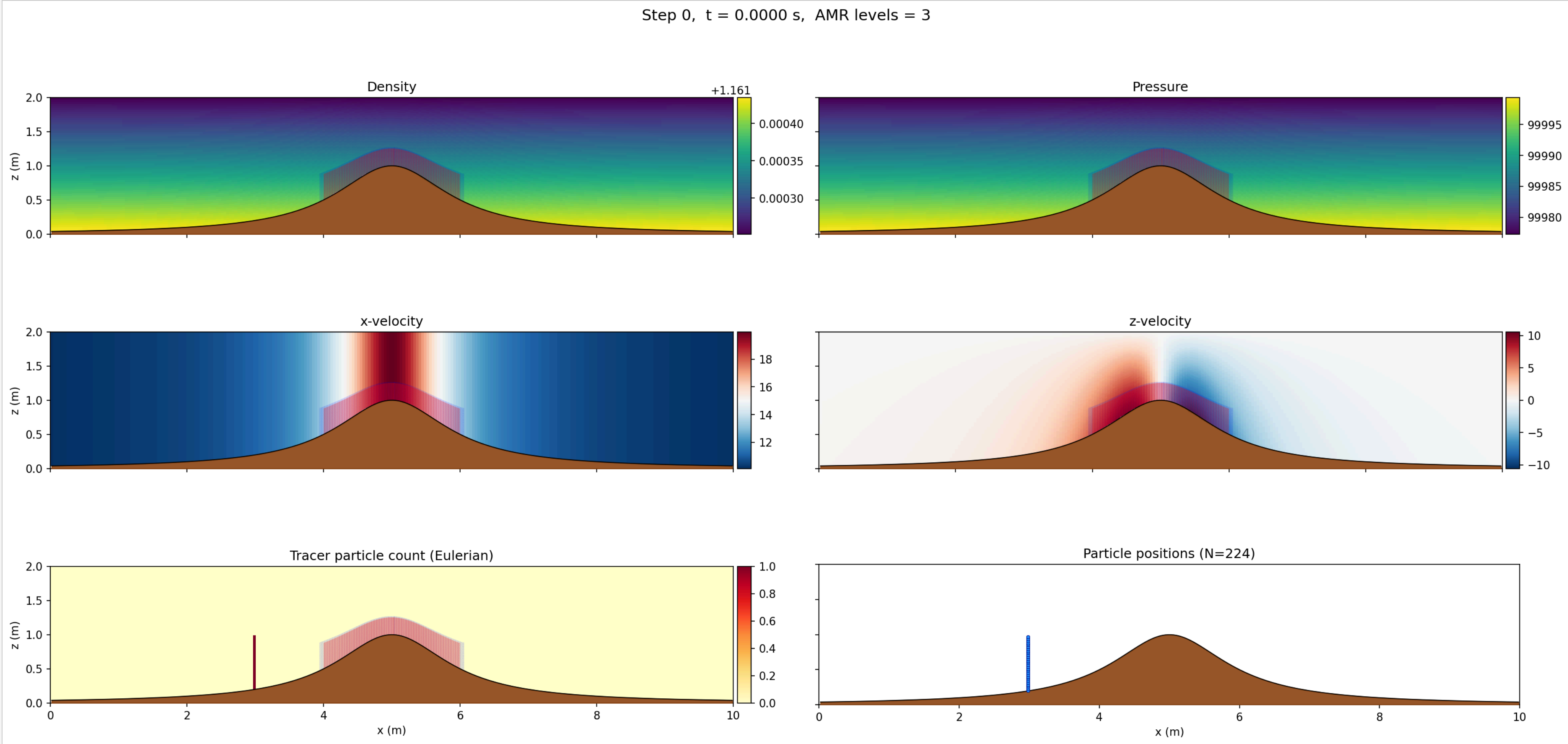
4. Terrain (WoA), AMR2, box_fullz

Input: [inputs_over_hill_AMR2_box_fullz](#)
FPE at coarse step 87 during L2 time integration. Two refined levels with terrain-following coordinates and full-z box coverage. Even with full-z coverage, 2 levels of refinement amplify the interpolation error enough to crash early.
Backtrace: signal caught in [powf64](#) (libm) called from [ERF_TI_utils.H:41](#) inside the slow RHS lambda of [advance_dycore](#) → [ERF_TI_slow_rhs_pre.H:26](#) . NaN/Inf from incorrect terrain C-F interpolation propagates into the RHS.



5. Terrain (WoA), AMR2, box_partialz

Input: `inputs_over_hill_AMR2_box_partialz`
FPE at coarse step 73 during L2 time integration. Earliest failure of all one-way coupling cases. Partial-z box + terrain + 2 refined levels is the most challenging configuration for the interpolation.
Backtrace: signal caught in `powf64` (libm) called from `ERF_TI_utils.H:41` inside the slow RHS lambda of `advance_dycore` → `ERF_TI_slow_rhs_pre.H:26` . Fails earliest because partial-z + terrain maximizes the z-offset error in the coarse-fine interpolation.



6. Terrain (WoA), AMR2, particlecount

Input: [inputs_over_hill_AMR2_particlecount](#)
Memory corruption ("corrupted size vs. prev_size") at coarse step 100. Dynamic tagging + terrain + 2 refined levels. Crashes during memory operations after the flow state has been corrupted by incorrect coarse-fine interpolation.
Backtrace: SIGABRT from heap corruption. Same root cause as case 3 but with an additional AMR level. The heap corruption originates from bad data produced by index-space interpolation kernels that assume uniform z-spacing in terrain-following coordinates.

